

EEN1095 Project Progress and Change Log

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Project Title	Design and Evaluation of Low-Sensitivity Digital Filters Implemented in Python Under Varying Numerical Precision Levels	Supervisor	Martin Collier
Implementation Period	Week 1 - Week X	Version Date	22/05/2026

Guidance: Maintain this as one **cumulative** document throughout EEN1095. Add a new entry every two weeks and do not remove earlier records. Progress should be recorded in a timely and concise manner. Where no change has occurred, this should be stated explicitly.

1. Project Overview

Original approved project title	Design and Evaluation of Low-Sensitivity Digital Filters Implemented in Python Under Varying Numerical Precision Levels
Original project objectives	Evaluate three IIR filter structures (Direct-Form, Ladder, WDF) and FIR filters across four Python precision levels (float16, float32, float64, mpmath) using five quantitative metrics to determine whether structure or precision is the primary robustness driver.
Original methodology proposed	Systematic multi-precision comparative evaluation in Python 3.11 using NumPy, SciPy, and mpmath. All filters designed via SciPy with transition bandwidth 0.02. Metrics computed: pole displacement, stability margin, sensitivity norm, frequency response deviation, roundoff noise.
Planned outputs / deliverables	Complete Python multi-precision evaluation framework; experimental results dataset; comparative analysis tables and figures; final project report (August 2026); open-source Python tool for precision-aware filter design.

2. Summary of Key Milestones

The table below is intended as a high-level summary. Use Section 3 for the detailed bi-weekly updates.

Period	Planned activity	Progress made	Outcome / status
Weeks 1-2	Begin Python multi-precision framework development. Set up NumPy, SciPy, and mpmath. Validate precision control for float16, float32, float64, and mpmath.	Python framework structure established. Precision control for float16, float32, float64, and mpmath confirmed working. Pilot filter design tested using SciPy.	In Progress — M3: Python framework validated at all 4 precision levels
Weeks 3-4	Complete Python framework validation. Implement direct-form IIR evaluation module. Run pilot experiments for order-10 and order-20 IIR designs.	Not yet reached. Scheduled for next period.	Upcoming
Weeks 5-6	Implement ladder filter Python module. Validate against direct-form baseline.	Not yet reached. Scheduled for June 2026.	Upcoming — M4: All filter structures implemented and validated
Weeks 7-8	Implement WDF Python module. Begin full IIR experimental evaluation across all 144 configurations.	Not yet reached. Scheduled for June 2026.	Upcoming — M5: Complete experimental dataset collected
Weeks 9-10	Complete full IIR and FIR experimental evaluation. Collect complete results dataset.	Not yet reached. Scheduled for June–July 2026.	Upcoming
Weeks 11-12	Perform cross-precision statistical analysis. Produce comparative tables and figures.	Not yet reached. Scheduled for July 2026.	Upcoming — M6: Analysis and comparative figures complete
Weeks 13-14	Write and submit final project report. Publish open-source Python framework.	Not yet reached. Scheduled for August 2026.	Upcoming — M7: Final report submitted

3. Detailed Progress Entries

Entry 1: Weeks 1-2

Main objectives for this period	Begin Python multi-precision framework development using NumPy, SciPy, and mpmath. Set up and validate precision control for float16, float32, float64, and mpmath. Run pilot filter design tests using SciPy.
Work completed	Python framework structure established. NumPy dtype selection confirmed working for float16, float32, and float64. mpmath precision control via mp.dps = 50 confirmed working. Pilot filter design tested using SciPy scipy.signal for Butterworth and elliptic IIR designs. SciPy SOS format validated for numerical accuracy in high-order IIR designs.
Key outputs produced	Python framework files: filter_design.py, core filter design and precision conversion module. metrics.py, five performance metrics module. experiment.py, experimental evaluation runner. Pilot results for order-10 and order-20 direct-form IIR designs across all four precision levels.
Problems encountered	No significant problems encountered. Minor issue with mpmath coefficient conversion from SciPy float64 arrays, resolved by converting each coefficient individually using mpmath.mpf(str(c)) to avoid floating-point representation errors during conversion.
Changes made to plan or methodology	No changes to plan or methodology at this stage. Framework implementation proceeding as planned.
Reasons for any changes	No changes required. Python framework structure and precision control confirmed working as originally planned.
Next steps	Weeks 3-4 — complete full Python framework validation across all four precision levels. Implement direct-form IIR evaluation module. Run pilot experiments for order-10 and order-20 IIR designs across all precision levels and record results.

Entry 2: Weeks 3-4

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]

Next steps	[Insert concise notes here]
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Entry 3: Weeks 5-6

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]
Next steps	[Insert concise notes here]

Entry 4: Weeks 7-8

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]
Next steps	[Insert concise notes here]

Entry 5: Weeks 9-10

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]

Next steps	[Insert concise notes here]
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Entry 6: Weeks 11-12

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]
Next steps	[Insert concise notes here]

Entry 7: Weeks 13-14

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]
Next steps	[Insert concise notes here]